Surinder Singh Chhabra

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Summary

ML engineer with experience building **end-to-end pipelines** spanning model training, graph-based retrieval, **LLM integration**, and **production deployment**. Currently completing an M.S. in Computational Science (Data Science) at SDSU, with active research in **RAG systems**, diffusion model benchmarking, and **explainable AI**. Proficient in PyTorch, LLaMA, XGBoost, SHAP, Docker, and MLflow.

Education

San Diego State University — San Diego, CA
Master's in Computational Science – Data Science — *Aug. 2024 – Present*

Vellore Institute of Technology — Vellore, India
Bachelor of Technology in Computer Science and Engineering — *Sept. 2020 – May 2024*

Work Experience

San Diego State University — San Diego, CA
Research Assistant – AstroRAG | Python, RAG, LLaMA-3.1, BM25, NetworkX, Groq — *May 2026 – Aug. 2026*

- Architecting a production-grade five-stage RAG system over a corpus of 408K arXiv papers, implementing **hybrid retrieval (BM25 + dense embeddings)** to improve recall over sparse-only baselines.
- Designing a graph-based reranking layer using four domain signals (concept similarity, bibliographic coupling, co-citation, domain hierarchy) to construct a weighted adjacency matrix and run **Personalized PageRank** for citation-aware relevance propagation.
- Integrating Groq-hosted **LLaMA-3.1-8B** as a cross-encoder reranker with graph-adjusted scoring, **reducing false positives in top-k retrieval** and improving response faithfulness over a BM25-only baseline.
- Implementing a structured **quality-assessment gate** (faithfulness + completeness + informativeness; threshold ≥ 0.75) with automatic re-selection fallback, preventing hallucinated or incomplete LLM outputs from reaching end users.

San Diego State University — San Diego, CA
Research Intern, CSRC Dept. | Python, PyTorch, TensorFlow, Diffusion Models, MLflow — *May 2025 – Jul. 2025*

- Implemented and trained **4 deep learning models** (LSTM, Transformer, Diffusion-TS, ST-GCN) for spatiotemporal forecasting under supervisor guidance, writing modular PyTorch code reviewed in weekly check-ins.
- Wrote **standardized training and evaluation scripts** to measure MAE, RMSE, and MAPE across models, achieving **MAPE $<$ 8%** on held-out test sets and **reducing inter-run variance by $\sim$15%**.
- Logged all experiments in **MLflow** (parameters, metrics, artifacts) across **20+ runs**, maintaining clean experiment records used directly by the supervising faculty for a **peer-reviewed publication (in submission)**.
- Adapted existing model code to a new dataset by reshaping input tensors and updating data loaders, gaining hands-on experience with **real-world data preprocessing and model debugging workflows**.

Projects

Churn Prediction & Deployment System | *Python, XGBoost, SHAP, FastAPI, Docker* — Aug. 2025 – Oct. 2025

- Framed churn prediction as a binary classification problem on **7K+ customer records**; engineered **20+ behavioral and usage features**, conducted EDA, and handled **class imbalance via SMOTE** before model training.
- Trained and compared Logistic Regression and XGBoost classifiers with 5-fold cross-validation; tuned XGBoost via grid search achieving **ROC-AUC 0.846 and F1 0.79**, **outperforming the logistic baseline by 12%**.
- Implemented a business-aligned decision threshold optimizer (threshold = 0.10) to maximize expected campaign profit; quantified ROI at $\approx$ **$37,884** and surfaced top churn drivers per customer using **SHAP TreeExplainer**.
- Packaged the trained model as a REST inference API using FastAPI, containerized with Docker, and deployed on AWS EC2 — enabling real-time churn scoring with **sub-100ms latency** for downstream CRM integration.
- Built a Streamlit monitoring dashboard exposing risk score distributions, threshold sensitivity curves, and SHAP waterfall plots, enabling non-technical stakeholders to interpret and act on model outputs directly.

Stock Forecasting – LSTM + Sentiment Fusion | *TensorFlow, Pandas, Oracle DB* — Jun. 2023 – Aug. 2023

- Designed a multimodal time-series dataset by engineering 12 OHLCV-derived technical indicators (RSI, MACD, Bollinger Bands, ATR) and fusing them with NLP sentiment scores from financial news APIs, timestamp-aligned to market trading windows.
- Built a sequence-to-one LSTM architecture in TensorFlow with configurable lookback windows; applied **Bayesian Optimization** over learning rate, sequence length, batch size, and hidden units — achieving **90%+ improvement in directional accuracy** vs. a moving-average baseline.
- Validated model generalization using **walk-forward time-series cross-validation** to prevent data leakage, ensuring evaluation reflected real trading conditions rather than in-sample overfitting.
- Automated daily data ingestion from Oracle Database into preprocessed tensor datasets, enabling scheduled retraining and eliminating manual data pipeline overhead across **2+ years of historical price data**.
- Analyzed feature importance via permutation testing to quantify the marginal lift of sentiment signals over price-only features, providing interpretable evidence that **NLP fusion improved prediction by $\sim$8%** on volatile trading days.

Technical Skills

Languages: Python, Java, SQL, C/C++, JavaScript, MATLAB, JAX
ML & DL: PyTorch, TensorFlow, XGBoost, Scikit-learn, SHAP, **Bayesian Optimization**, Diffusion Models, LSTMs
LLM & RAG: LLaMA-3.1, Groq API, BM25, **Personalized PageRank**, K-means Retrieval, Prompt Engineering
MLOps: MLflow, Docker, FastAPI, Streamlit, AWS EC2/S3, GCP Vertex AI, Git
Data: Pandas, NumPy, NetworkX, Matplotlib, Tableau, Oracle DB